

25. The harmonic mean of two real numbers x and y equals $2xy/(x+y)$. By computing the harmonic and geometric means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

Solution: ① $x=8$ $y=2$ $G.M. = \sqrt{xy} = \sqrt{16} = 4$ $H.M. = \frac{2 \times 8 \times 2}{8+2} = 3.2$

② $x=5$ $y=5$ $G.M. = \sqrt{xy} = 5$ $H.M. = \frac{2 \times 5 \times 5}{5+5} = 5$

③ $x=16$ $y=4$ $G.M. = \sqrt{64} = 8$ $H.M. = \frac{2 \times 16 \times 4}{16+4} = 6.4$

④ $x=27$ $y=3$ $G.M. = \sqrt{81} = 9$ $H.M. = \frac{2 \times 27 \times 3}{27+3} = \frac{162}{30} = 5.4$

Claim: For any two arbitrary real numbers x and y , $G.M. = \sqrt{xy}$

$H.M. = \frac{2xy}{(x+y)}$

Proof: Suppose for the sake of contradiction, $\sqrt{xy} < \frac{2xy}{(x+y)}$. Squaring both sides, $xy < \frac{(2xy)^2}{(x+y)^2} \Rightarrow xy < \frac{4x^2y^2}{x^2+2xy+y^2}$

$\Rightarrow xy(x+y)^2 < 4xy \cdot xy$. If xy is non-negative, then:
 $(x+y)^2 < 4xy \Rightarrow x^2+2xy+y^2 < 4xy$

$\Rightarrow x^2-2xy+y^2 < 0 \Rightarrow (x-y)^2 < 0$

$(x-y)$ is a real no. $\therefore (x-y)^2 \geq 0$ (A Fundamental Property of Real Numbers)

This is a contradiction.

$\therefore$ For any two arbitrary real numbers x and y , $\boxed{\sqrt{xy} \geq \frac{2xy}{(x+y)}}$ $\square$